

**CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated
for lumbosacral transitional anatomy**

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Purpose: A vertebra at the lumbosacral junction is named by counting. The gold standard counts caudally from C2 on whole-spine imaging¹; however, a lumbar case is planned on a lumbar study^{2,3}, T12 to S1, without C2. Abdominopelvic computed tomography (CT) holds that span plus the lowest ribs and pelvis, standing in for the preoperative view. Where a lumbosacral transitional vertebra (LSTV) alters the count, the local anatomy is ambiguous: four rib-free vertebrae may be an L1 with a lumbar rib or an L5 assimilated to the sacrum, and six may be a sixth lumbar vertebra, a T12 with aplastic ribs, or a lumbarized S1⁴. CTSpinoPelvic1K asks whether local morphology resolves that ambiguity without the count. Two public label sets already covered this imaging, CTSpine1K’s vertebrae and CTPelvic1K’s pelvis, on the same colonography patients, never joined; this release joins them on one series and annotates the bones neither had. It provides 802 CT records with per-level ribs and femora, lumbar levels anchored on the lowest rib-bearing vertebra and S1, and classes for L6, T13, a separate S1 and lumbar ribs, so anomalies are recorded as themselves.

Acquisition and Validation Methods: Records pair CTSpine1K⁵ and CTPelvic1K⁶ labels on each patient’s bone-richest acquisition under a VerSe-native scheme. Validation: geometric invariants (802/802 pass); rib–vertebra incidence across 5,749 evaluable ribs (0.035% offset); and spinopelvic measures matching published values.

Data Format and Usage Notes: NIfTI image/label pairs with patient-grouped LSTV-stratified five-fold splits and a reference loader; archived at <https://doi.org/10.5281/zenodo.22139642>.

Potential Applications: Classifying a vertebra from local features; updating cadaveric morphometry; spinopelvic assessment; opportunistic screening; and, absent a public preoperative lumbar CT cohort, planning research (377 records prone). *Limitations:* thoracic ground truth is field-of-view limited; postural angles recumbent; no held-out test set; ribs are triaged-review pseudolabels; Castellvi grades are two-reader consensus.

6 I. INTRODUCTION

7 The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, and
8 a vertebra is identified by where it falls in that sequence. Transitional variants sit at the
9 two ends of the lumbar column, the thoracolumbar transitional vertebra (TLTV) above and
10 the lumbosacral transitional vertebra (LSTV) below, and they disturb identification in two
11 ways at once. They change what the border vertebra and its ribs *look like*: a thoracolumbar
12 border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral
13 one may be partly or completely assimilated to the sacrum (*sacralization*), with an anteriorly
14 wedged body, a reduced disc beneath it and hypoplastic facet joints, or conversely separated
15 from it (*lumbarization*), with a squared body, lumbar-type facets and a full-height disc⁴.
16 They also change *how many* vertebrae each region contains: eleven or thirteen thoracic,
17 four or six lumbar.

18 Either change alone would be tractable. Together they make it difficult to state defini-
19 tively what any vertebra at these borders *is*. Four rib-free vertebrae above the sacrum may
20 mean an L1 bearing a lumbar rib or an L5 assimilated to it; six may mean a true sixth
21 lumbar vertebra, a T12 whose ribs are aplastic, or an S1 lumbarized away from the sacrum.
22 Each set yields the same count, and the morphology that would separate the readings is not
23 decisive to a human reader in every case. Whether it differs enough for a learned classifier
24 to separate them, or at least to assign each reading a probability, is the question this release
25 is built to support. Absent that, what remains is the count itself: the gold standard for
26 numeration in transitional anatomy is whole-spine imaging, counting caudally from C2¹,
27 which fixes how many vertebrae and how many rib pairs the column holds, and no spine-
28 limited acquisition can supply either count. LSTV is common, reported between 4% and
29 30% depending on definition, and wrong-level surgery is its most serious consequence.

30 Existing public collections cannot address this, and size is not the reason (Table I). Spine
31 datasets omit the pelvis; pelvic datasets do not number the vertebrae; TotalSegmentator,
32 the whole-body scheme, has no class for a sixth lumbar vertebra or for a rib on a lumbar
33 vertebra; and VerSe, the one collection with an L6 class, stops at the sacrum and carries
34 no ribs. No public collection can therefore record a six-lumbar spine together with both
35 of its borders, however good the segmentation, and a segmenter trained on them inherits
36 the gap: faced with an enumeration anomaly it must shift the whole column by a level or

absorb a vertebra into its neighbor. The size of that problem has now been measured on CT, where prior labeling methods assigned every vertebra correctly in 77% of subjects and an anomaly-aware extension of SPINEPS⁷ raised that to 99%⁸. Its weights label L6 and T13 on CT, but its 1,536 training scans are in-house and unreleased: the capability exists as weights, not as data.

The intraoperative side of the problem has prior art of its own. LevelCheck registers the intraoperative radiograph to the preoperative CT and projects the CT’s vertebral labels onto it, localizing the target level in a limited field of view to within a few percent of cases in clinical use^{9–11}. The labels it projects are placed by hand on the CT, by a skilled operator and verified by the surgeon⁹: the method assumes a correctly labeled CT and does not supply one. Automating that step, and getting it right precisely where the anatomy is anomalous, is what a dataset carrying the classes those anomalies produce is for.

CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and gives a class to each structure an enumeration anomaly produces: a sixth lumbar vertebra, and a rib borne by a lumbar vertebra. A scheme without those classes lacks a class vocabulary expressive enough to describe anatomic anomalies.

The project began as a surgeon’s question: whether a vertebra can be named from the anatomy in front of the reader, on the images a lumbar case is actually planned on: by guideline those are lumbar studies^{2,3,12}, T12 to S1, and whole-spine coverage is reserved for trauma with an identified injury¹³ and for deformity radiographs. The cohort suits that question: an abdominopelvic CT runs from the diaphragm to the pelvic floor, so it holds the T12-to-S1 span of the planning study together with the lowest ribs and the whole pelvis, and it lacks C2 exactly as that study does. Every record comes from one prospective trial protocol, ACRIN 6664^{14,15}: 802 patients aged 50 and over, scanned supine and prone on multidetector CT at 15 centers, on five scanner vendors and nine models, with manufacturer, model and reconstruction kernel recorded per record. To our knowledge it is the largest spine-and-pelvis annotated CT cohort acquired under a single protocol; the comparators pool several collections (CTSpine1K four, CTPelvic1K seven), are multi-site by design (VerSe), or are routine clinical scans under many protocols (TotalSegmentator). With the protocol held fixed, scanner effects on a model become measurable rather than confounded. The price is the field of view: abdominal, so the thoracic spine enters from below and C2 is never present (Sec. V).

69 The gap was in plain sight. CTPelvic1K⁶ and CTSpine1K⁵, released months apart by
70 an overlapping author group, each annotated the COLONOG collection under radiologist
71 supervision, one the pelvis and the other the vertebrae, on the same patients, and no one
72 joined them, although the join needs no new imaging and no new radiologist. It is not a
73 file merge: every patient was scanned prone and supine and neither source recorded which
74 series it drew on, so each annotation had to be traced back to its own series (Sec. II.A).
75 Done once, that crosswalk yields a spine-and-pelvis frame at 802 patients for the cost of
76 the crosswalk, and the bones neither set had, ribs, femora, S1 and hardware, could then be
77 annotated against it rather than from scratch. Two of the most-used public bone annotation
78 sets were halves of one dataset; this release is the whole.

79 *Where the material is.* Everything described here is v8 of one archived dataset, <https://doi.org/10.5281/zenodo.22293878>;
80 cite the concept DOI <https://doi.org/10.5281/zenodo.22139642>, which resolves to the current version (Sec. III).
81

82 I.A. Vertebra labeling when the scan cannot settle the count

83 *The limitation, stated plainly.* No scan in this corpus contains C2, so the conventional
84 count cannot be performed: a property of abdominal imaging, not of the annotation. A
85 thirteenth thoracic vertebra, a lumbar rib and a *stump rib* (a hypoplastic twelfth rib, a
86 cardinal indicator of a thoracolumbar transitional vertebra) leave the same number of rib-
87 free vertebrae between the two borders of the lumbar spine, the last rib-bearing vertebra
88 and the sacrum.

89 *What the release does in addition.* Every vertebra carries the identifier its radiologist-
90 sourced annotation assigned, corrected where the source was wrong; those labels are the
91 ground truth. Alongside them, every released measurement is also expressed against the
92 two landmarks present in essentially every abdominal scan, the sacrum below and the lowest
93 rib-bearing vertebra above, so that the measurement is positional rather than nominal: a
94 vertebra by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its
95 length as a fraction of the rib above it, a transverse process by its span and its distance from
96 the ala. None of these changes according to whether a reader would have said L5 or S1, so
97 cases that disagree about the name remain comparable, and where the junction is in view
98 the count is determined and the two descriptions agree.

TABLE I Public CT collections at the lumbosacral junction. Black, class present; outlined, graded only to exclude it²¹.

Collection	Scans	Count	L6	Sacrum	S1	Pelvis	Ribs	L. rib	Femora	Grade
CTSpine1K ⁵	1,005									
CTPelvic1K ⁶	1,184									
VerSe ²²	374									
RibSeg v2 ²³	660									
TotalSegmentator ²⁴	1,204									
CTSpinoPelvic1K	802									

99 *But the vertebrae themselves are known to differ.* Counting is not the only route to an
100 identity. Level-specific morphometry is long established: pedicle width and height change
101 systematically from the thoracic to the lumbar spine,¹⁶ vertebral body, endplate and canal
102 dimensions differ by level,^{17–19} and the transverse process and the iliolumbar ligament mark
103 the last lumbar vertebra independently of any count.^{4,20} If those differences hold at the
104 boundary itself, the phenotype is decidable from the vertebra in front of the reader without
105 any global count. On this corpus it does hold: separating T12 from L1 on shape alone, with
106 each patient’s own size divided out and cross-validation grouped by case, gives an area under
107 the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. Size is removed because raw
108 millimeters separate thoracic from lumbar trivially and teach nothing about the junction,
109 where the two are nearly the same size.

110 That is reported as a property of the released measurements rather than as a method:
111 the information needed to settle these variants is present locally.

112 II. ACQUISITION AND VALIDATION METHODS

113 II.A. Sources, and why a crosswalk was necessary

114 Both sources draw part of their imaging from the TCIA CT colonography collection^{14,15}.
115 CTSpine1K’s 1,005 volumes come from four collections and CTPelvic1K’s 1,184 from seven;
116 COLONOG is the only collection common to both and the only one acquired under a single
117 protocol, so this release is its COLONOG subset: 782 patients with a CTSpine1K vertebral

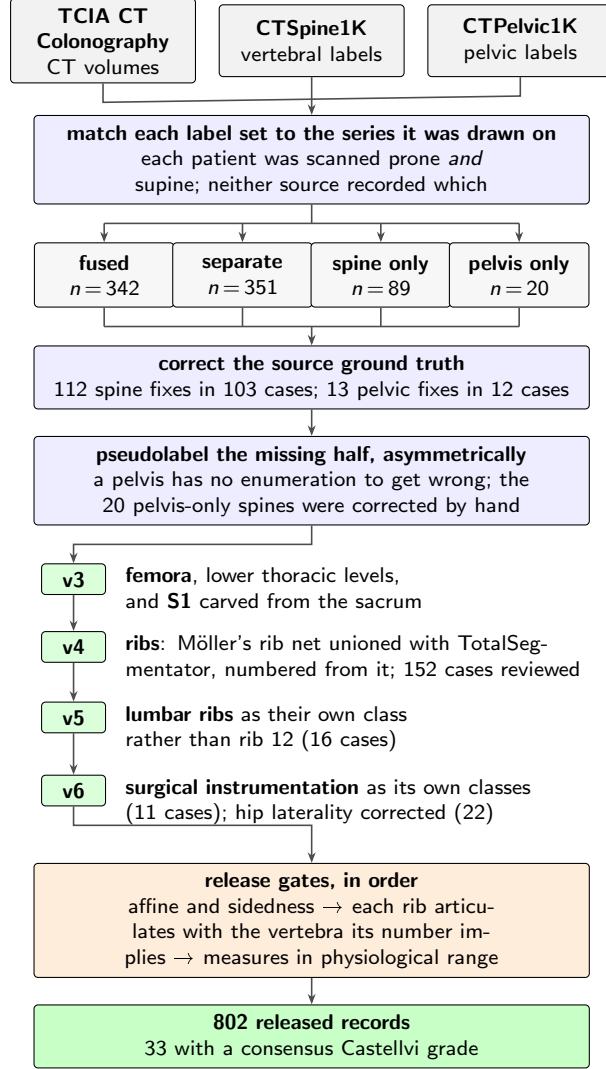


FIG. 1 How the dataset was built. n counts scans, or corrections, at each step.

118 annotation and 20 with a CTPelvic1K pelvic annotation only, 802 in all. Neither source
119 released the mapping from an annotation to the specific CT series it was drawn on.

120 That mapping matters because every patient in CT COLONOGRAPHY was scanned
121 *twice*, prone and supine, and an annotation carries a patient identifier and nothing finer.
122 Turning a patient from supine to prone alters lumbar lordosis and the alignment of each
123 segment against the next, so outlines drawn on one series are the wrong *shape* for the other
124 and no rigid realignment recovers them. The crosswalk (Fig. 1) therefore resolves each
125 annotation to a patient and then, within that patient's volumes, to the series whose bone
126 *agrees* with it: candidates are thresholded at bone attenuation and scored by overlap with
127 the mask.

II.B. Fused and split records

CTSpine1K and CTPelvic1K each annotated a patient independently, so their labels are not guaranteed to come from the same CT series (Fig. 1). This turned out to matter for 351 patients, nearly half the cohort: the two collections had labeled different acquisitions of the same person, one prone and one supine, with neither recording which. Because posture changes spinal alignment, a spine label from one acquisition cannot simply be paired with a pelvic label from the other. These 351 records are released but held out of any analysis that assumes a single posture; two labeled postures of one patient are a resource for studying alignment, not a defect.

II.C. Densification, and why it is asymmetric

A corpus in which half the records lack a pelvis is not a spinopelvic dataset, so the missing structures were pseudolabeled, asymmetrically (Fig. 1). Pelvis onto spine-only records is the safe direction: the sacrum and innominate bones carry no enumeration to get wrong, and quality is reported as held-out performance on the *pelvic_only* records, withheld from training for that purpose. The reverse direction carries the whole problem this dataset addresses, since a pseudolabeled spine must commit to a count and on a transitional vertebra commits to the commoner reading; those 20 records were corrected by hand, tractable at that scale.

II.D. Ribs

No public rib annotation covers this cohort, and the two available automatic sources fail in complementary ways.

Möller’s binary rib network²⁵ reports high accuracy, and its authors show that stump ribs can be classified from a partial field of view; that result concerns the morphological features computed on a rib once segmented. On this corpus the segmentation itself failed on ribs that were *partially outside* the field of view: a rib crossing the boundary of an abdominal acquisition was liable to be missed altogether rather than segmented to the edge. Its authors do not report this: their evaluation excluded subjects whose last rib was not visible, so it is recorded here as an observation from this cohort. In a colonography cohort the lower ribs are routinely clipped, and those are the ribs the count depends on.

156 TotalSegmentator²⁴ does segment clipped ribs and supplies the per-rib numbering that a
 157 binary network cannot. Its characteristic failure is at the other end of the bone: the most
 158 medial portion of the rib (roughly the last tenth to sixth approaching the costovertebral
 159 joint) is frequently absent. A rib truncated at its medial end never reaches the vertebra
 160 it articulates with, so the rib-vertebra incidence check (the quality-control gate on the rib
 161 class labels, which matches each rib to the vertebra its number implies) has nothing to test.

162 The two were therefore unioned (Fig. 1), TotalSegmentator supplying the numbering and
 163 the medial-clipped cases, Möller the detection where its output is more complete.

164 The result is a pseudolabel whose review was triaged rather than exhaustive. A label-
 165 based rule flagged every record in which one rib bone carried more than one identifier, or in
 166 which a rib failed to reach the vertebra it articulates with, since a single bone carrying two
 167 numbers is a numbering error by construction, whatever the anatomy. The rule selected 152
 168 records; 148 were reviewed and corrected, and four were not and are identified in the release.
 169 Reviewers were medical students working through a student research consortium²⁶, trained
 170 against a written labeling protocol, with every correction recorded against the annotator
 171 who made it. Records passing the rule were not individually inspected, so the layer only
 172 guarantees freedom from the failure modes the rule detects, no more. The residual rib-
 173 vertebra offsets reported in Sec. II.G are the independent check on what the rule missed.

174 II.E. Label scheme and counting anchors

175 The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12
 176 = 8–19, L1–L6 = 20–25, sacrum 26), and every non-VerSe structure takes a fixed identifier
 177 above that range: S1 carved from the sacrum (29), hips (30–31), femora (32–33), ribs (34–45
 178 left, 46–57 right), lumbar ribs (74–75) and surgical hardware (76–82). Identifiers 58–73 are
 179 unassigned: a soft-tissue block once reserved there was never populated and is retired.

180 Two anchors make the count decidable, and Fig. 2 shows both: the lowest rib-bearing
 181 vertebra above, **S1** carved from the sacrum below. Each earns its place by what it supports
 182 rather than by being an endpoint. S1 supplies the sacral endplate landmark that sacral
 183 slope and pelvic incidence are measured from, and with both brackets present L5 and L6
 184 are distinguishable where a spine-limited view alone would not be.

185 *How S1 is cut.* The sacrum’s outer boundary is the source annotation and is never altered;

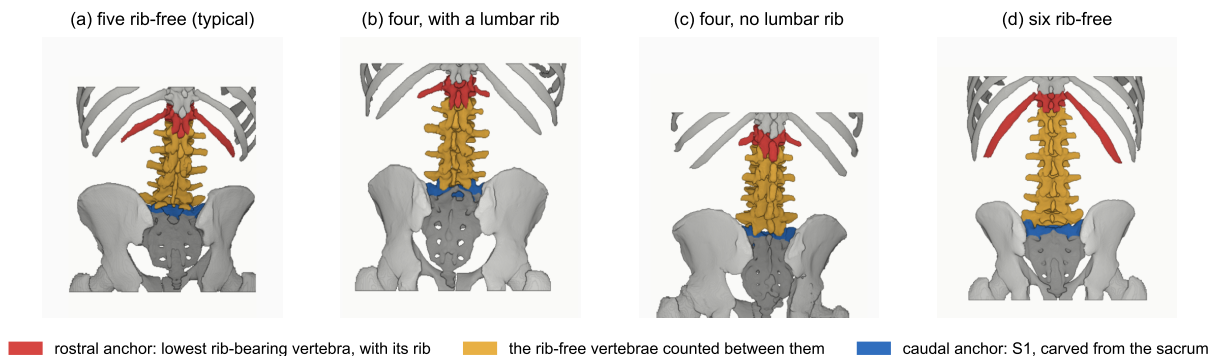


FIG. 2 The two anchors, from the released labels. Red, lowest rib-bearing vertebra and its rib; blue, S1; yellow, the rib-free vertebrae between. (b) and (c) both count four.

S1 is the cranial part of that same bone, separated by a plane. The plane's normal is the sacrum's own cranio-caudal axis, made orthogonal to a left-right axis measured directly rather than assumed, so as to remove the patient's rotation within the scanner. That rotation is not negligible: median 4.5° , maximum 16.1° , and 508 of 801 records above 3° . The cut preserves the sub-division volume of the preceding release, changing the orientation of the S1/S2 boundary and not how much sacrum is called S1; per-record geometry ships with the archive.

Neither anchor is novel on its own: TotalSegmentator²⁴ carries a separate S1 class and per-level ribs, and VerSe²² carries L6, whose identifier this release keeps, so its L6 is VerSe's.

A rib on a lumbar body receives its own class, and this is the one class here with no published counterpart. A lumbar rib and a stump rib are the same object under two counts, so assigning it a number would commit the label to one reading of an enumeration anomaly. A scheme that numbers every rib 1–12 has nowhere to put a thirteenth: the annotator must either call it rib 12, asserting the vertebra beneath it is thoracic, the very question at issue, or discard it. TotalSegmentator carries ribs 1–12 per side and no such class, and VerSe's T13 is a vertebra rather than a rib. This scheme can record both: VerSe's T13 (28) for a thoracic-type thirteenth vertebra with a true rib, and 74/75 for a lumbar-type body bearing a rudimentary one, so either reading is recorded as itself. No record here carries a T13; the sixteen lumbar-rib cases were read as lumbar-type bodies. Where a border vertebra could not be settled by its readers the label stands as read, and reclassifying those records with a classifier trained on local vertebral morphology (Sec. V) is the intended path for a later release.

Figure 2 shows why the interval is the primitive and the label is derived. Among the records with four rib-free vertebrae, some reach that count through a lumbar rib and some without one; the two are indistinguishable by the count alone and are distinguished here only because rib presence is itself labeled.

Sixteen records carry a lumbar rib: thirteen bilateral and three unilateral, all on the right, a count too small to support any statement about side preference.

Eighteen records carry an L6. Fourteen are labeled lumbarization, two sacralization, one semi-sacralization and one normal, which is worth stating because it shows the two annotation axes disagreeing in the open: a record can carry six lumbar-type vertebrae and still be read as a sacralization, or as unremarkable, depending on where the reader started the count.

Both counts are derived from the released label volumes, and so are the manifest fields that report them. That is a correction: through v5, `has_l6` was true for one record that contains no L6 and false for all eighteen that do, `has_lumbar_rib` was false everywhere, and `n_lumbar_labels` read zero in 799 of 802 records. The fields are rebuilt for this release by counting identifiers 25 and 74–75 in each volume, with `lumbar_rib_side` added alongside them.

II.F. Computational tools

Access. Every step is performed by open-source code archived with the dataset, together with the reference loader, the label scheme and the quality-control scripts that generated the tables in this manuscript.

Versions and parameters. Segmentation of femora, the sacral sub-division and the per-level rib numbering used TotalSegmentator²⁴ (`total` task, release 2.x) on a single graphics processing unit. The binary rib network is Möller’s²⁵ released weights, applied through the nnU-Net v2 framework²⁷. The pelvic completion for records whose pelvis was absent used a five-fold nnU-Net v2 ensemble, applied out-of-fold: each record is completed by the fold that never trained on it, so no record is completed by a model that has seen it. The five fold checkpoints, with their nnU-Net plans and dataset descriptor, are released alongside the data (the repository URL identifies the authors; it is on the title page and will be restored here for the camera-ready version). Image handling throughout used NiBabel and SciPy; no

image is resampled or reoriented when a label is written, so every label shares its image's grid and affine exactly.

Generative artificial intelligence. A large language model (Claude, Anthropic) assisted with drafting and editing the manuscript text and with writing the analysis and figure code, all under author review. It was not used to generate, annotate or interpret imaging data; every number reported here is computed by the released code from the released volumes.

II.G. Validation

Validation is layered: measurements on a corpus that has not established internal consistency do not look broken, they look finished.

Geometric invariants. Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left-right axis read from the affine rather than assumed. The check exists because a renumbering pass once wrote a reoriented array under a canonical affine and transposed a label away from its image, a failure invisible downstream.

The check compares *each sided pair separately*: testing the ribs alone passed all 802 records while four carried `left.hip` on the patient's right, and pooling all sided structures into one test still missed one of those four, because a large correctly-sided structure masks a smaller transposed one. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

Rib-vertebra incidence. Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs, 5,788 are not evaluable because the expected vertebra lies outside the field of view and 11 have no vertebra within the anchor distance; of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is misnumbered. The denominator is stated because quoting the two offsets against all 11,548 ribs would halve the rate by counting ribs the check never examined. Both offsets are field-of-view truncations on individually characterized cases, one at +1 level and one at -1, reported rather than suppressed.

The same check carries an invariant that reads as a null result. Of the 5,749 evaluable ribs, *zero* are assigned to a lumbar vertebra, not because no lumbar ribs exist, since 16

TABLE II Derived measures against a standing reference ($n = 260$)²⁸; this cohort is supine.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	54.7 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis (supine)	52.7°	43 ± 11.2 (standing)
PI – LL mismatch	2.6°	centered near 0

records carry one, but because such a rib takes class 74 or 75 and never enters the numbered set the test examines. A numbered rib landing on a lumbar vertebra would be a counting error of exactly the kind this dataset exists to expose.

Agreement with published values. Derived spinopelvic measures are compared against Vialle *et al.*²⁸ (Table II). Pelvic incidence is the strongest of these checks because it is a morphological property of the pelvis rather than a posture, and so is directly comparable to a standing reference cohort.

II.H. Castellvi typing

Every record whose vertebral count departs from five carries a Castellvi grade²⁹ in the released manifest: 33 cases, typed I–IV with the *a/b* unilateral–bilateral qualifier. The grade and the count are *different axes*: grade IIIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a normal count of five.

Two radiology resident physicians graded every case independently and resolved their six disagreements by consensus; both reads and the consensus are in the manifest. The distinction the classification turns on, bony fusion against articulation without it⁴, is the one carrying clinical weight, and it is where the disagreements lay: four of the six were III against II.

II.I. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here because an iatrogenic fusion is indistinguishable from a congenital one to a distance

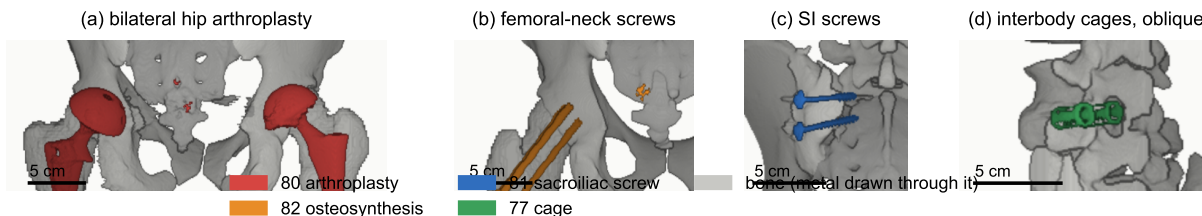


FIG. 3 Instrumentation in the release, one exemplar each, drawn through bone. (a) Hip arthroplasty, 80, $n = 8$. (b) Femoral-neck screws, 82, $n = 1$. (c) Sacroiliac screws, 81, $n = 1$. (d) Interbody cages, 77, $n = 1$. Bar, 5 cm.

measurement: a cage-bridged interspace reads as “no gap” exactly as a fused transitional vertebra does.

II.I.0.a. Detection over-calls by a factor of five. A 2500 HU threshold inside a shell around the labeled skeleton flags 52 of the 802 records. One of the two radiology resident physicians read all 52 and confirmed 11, rejecting 41 (79%) as contrast, calcification and reconstruction artefact. Attenuation does not separate the two groups (both saturate); site and volume do, every confirmed implant at $2,586 \text{ mm}^3$ or above and every rejection at $1,768 \text{ mm}^3$ or below.

II.I.0.b. The subtype block had to be extended. Identifiers 76–79 name spinal instrumentation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds, so **80 arthroplasty**, **81 sacroiliac screw** and **82 osteosynthesis** were added: osteosynthesis holds parts of one bone together where arthroplasty replaces a joint, and a shape rule calls a femoral stem a rod. The confirmed cases are eight arthroplasties, one osteosynthesis, one sacroiliac fixation and one pair of interbody cages (Fig. 3).

II.I.0.c. Metal outranks bone, and that is a subtraction. In every confirmed case the implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than adding them. In eight of these cases the femoral head that pelvic incidence and tilt are measured from is an implant.

III. DATA FORMAT AND USAGE NOTES

Each record is a gzipped NIfTI image/label pair sharing an identical 4×4 affine, canonicalized to PIR, requiring no resampling. Masks are NIfTI rather than DICOM, which is

TABLE III Data types and metadata, and the records populating each, from the released manifest.

Field	Records	Available
Identifier, image path, label path, configuration, match type	802	100.0%
Patient position, tube potential, section thickness, kernel	802	100.0%
Transitional label and class	802	100.0%
Spine and pelvic annotation origin	802	100.0%
Instrumentation and partial-annotation flags	802	100.0%
Image/label alignment check	802	100.0%
Spine series identifier; bone-fraction check	782	97.5%
Scanner manufacturer and model	768	95.8%
Sex	749	93.4%
Age	709	88.4%
Pelvic series identifier	713	88.9%
Castellvi grade (consensus of two readers)	33	4.1%

309 what volumetric segmentation tooling expects.

310 Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the
311 data as `splits_5fold.json`. The five validation folds partition all 802 records, so the
312 release provides cross-validation and *not* a held-out test set; users reporting a single held-
313 out number should carve one out and state which records it holds.

314 Stratification is on the transitional subtype rather than a binary LSTV flag: every val-
315 idation fold carries three sacralization and three lumbarization records. With 15 of each
316 across 802 that is the most stratification can guarantee, and it is why a fold-level metric on
317 the rare classes carries an error bar far wider than its own decimal places.

318 Table III lists the key data types and metadata against the fraction of records for which
319 each is populated.

320 The archive of record is Zenodo (<https://doi.org/10.5281/zenodo.22139642>). Code
321 and documentation are on GitHub at the tagged release commit; any model-hub copy is a
322 convenience mirror, not the archive.

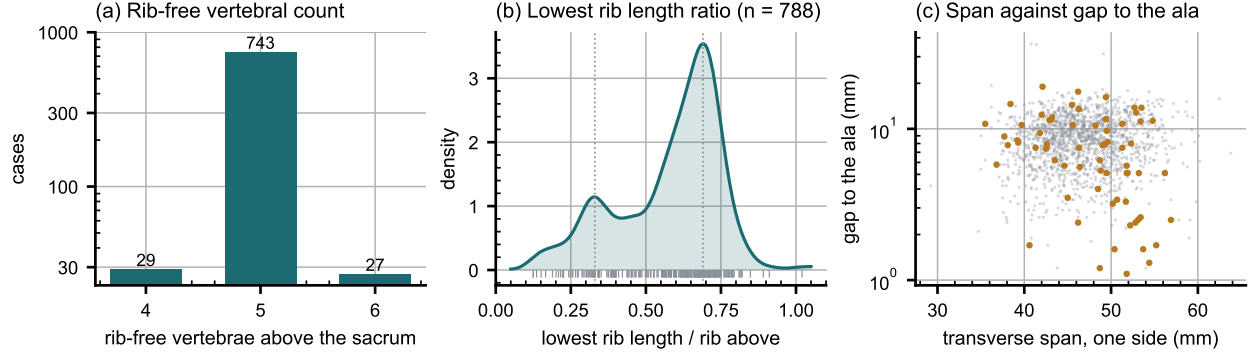


FIG. 4 Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b) Lowest rib length over the rib above. (c) Lowest lumbar transverse span against its gap to the ala; transitional labels marked.

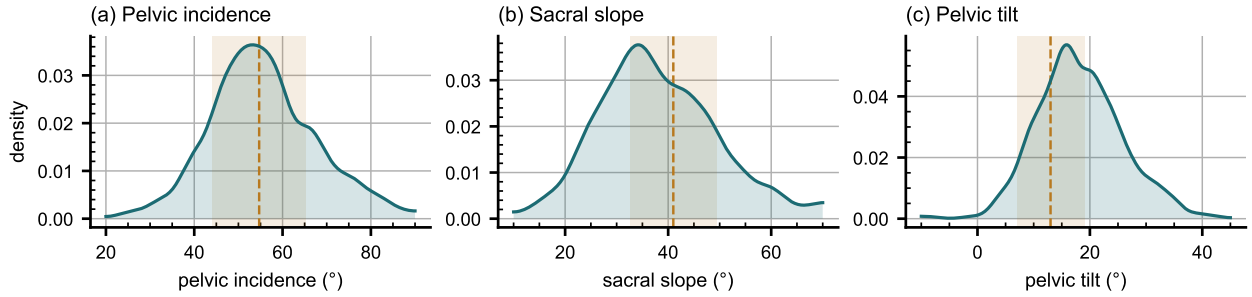


FIG. 5 Spinopelvic measures against standing reference bands (mean $\pm$ SD)²⁸.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM’s *other* value and 53 no sex field. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals; the eleven records outside the female/male dichotomy are excluded from sex-stratified measures rather than folded into “unrecorded”. The lower age bound is the screening guideline, not a selection applied here.

Count-free measures (Fig. 4). Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal.

Level gradients and spinopelvic measures (Fig. 5). Vertebral dimensions increase caudally; pelvic incidence does not vary with age while pelvic tilt does, the signature of a pelvis retroverting as a spine ages.

336 *Opportunistic measures.* Every scan carries a vertebral bone density measurement at no
337 additional dose.³⁰ L1 trabecular attenuation is reported at the published standard site.

338 V. POTENTIAL APPLICATIONS AND LIMITATIONS

339 The principal use is the *spinopelvic interface*. Spine and pelvis carry radiologist-sourced
340 annotation on one coordinate frame in 802 records, making sacral morphology measurable
341 against the lumbar column rather than in isolation: the sacral endplate and its slope, the
342 alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry
343 a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and
344 femoral heads rather than from landmarks placed on a radiograph, so it is reproducible from
345 the labels themselves.

346 The transitional layer serves that use rather than competing with it: no trajectory can
347 be planned across a junction whose segments cannot be named.

348 *A proxy for the preoperative view.* No public preoperative lumbar CT cohort exists that
349 we know of, and VerSe, the largest public spine CT collection, lacks the sacrum and pelvis
350 (Table I). This release is therefore the closest available stand-in for the view a lumbar case
351 is planned on: the T12-to-S1 span with the lowest ribs and the pelvis; 377 of 802 records
352 acquired prone, the position of posterior lumbar surgery; and classes for the anomalies that
353 make level identification fail. It is a screening population at colonography dose, reconstructed
354 with standard soft-tissue kernels rather than the bone kernel of a surgical CT, and not a
355 surgical series: it supports method development for level identification and instrumentation
356 geometry, not validation of a planning decision.

357 V.A. Three uses the released measurements support

358 *Naming a level from its own geometry.* Wrong-level spine surgery runs at roughly one in
359 3,110 spinal procedures, and the cause most often cited is the transitional anatomy this co-
360 hort was assembled around.³¹ Per-level body height, canal width, endplate width, transverse-
361 process span and wedge ratio are released for T11–L5 across all 802 records, and because
362 the labels were assigned against the twelfth rib rather than by enumeration, a model trained
363 on them is not learning the convention that fails.

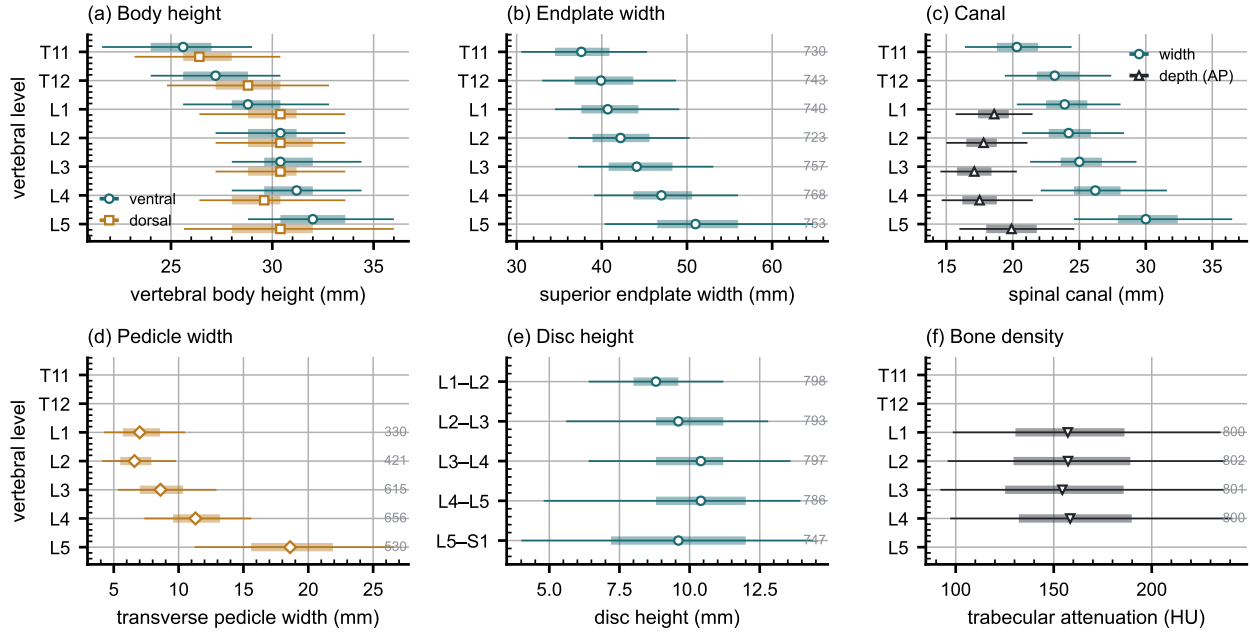


FIG. 6 Morphometry by level: median, interquartile range, 5th–95th percentile, n at right. (a) Body height. (b) Endplate width. (c) Canal. (d) Pedicle width. (e) Disc height. (f) Trabecular attenuation.

Reference morphometry with its spread. The values a surgeon consults for vertebral body height, canal dimensions and pedicle width come from cadaveric series of a few dozen specimens, plotted as single values with no indication of spread.¹⁹ They cannot tell a reader whether the patient in front of them is unusual, because they never show what usual looks like. The same measures are given here from 802 records with the distribution attached (Fig. 6), reproducing two properties of the classical figures (body height crosses over, dorsal exceeding ventral at T11–T12 and reversing by L4–L5, and pedicle width widens steeply into the lower lumbar spine) and adding the width of each distribution, which is the part needed to judge an individual.

Patient-specific models. Planning is moving toward patient-specific biomechanical models rather than population norms.³² Such a model needs spine, pelvis and femoral heads in one frame, which is this release’s construction; in 351 patients two postures give a within-patient change in alignment against which a predicted postural response can be checked.

Its limitations are stated at the level of detail a reader would need to catch them independently.

Coverage. Thoracic ground truth is field-of-view limited (Fig. 7) and does not extend to

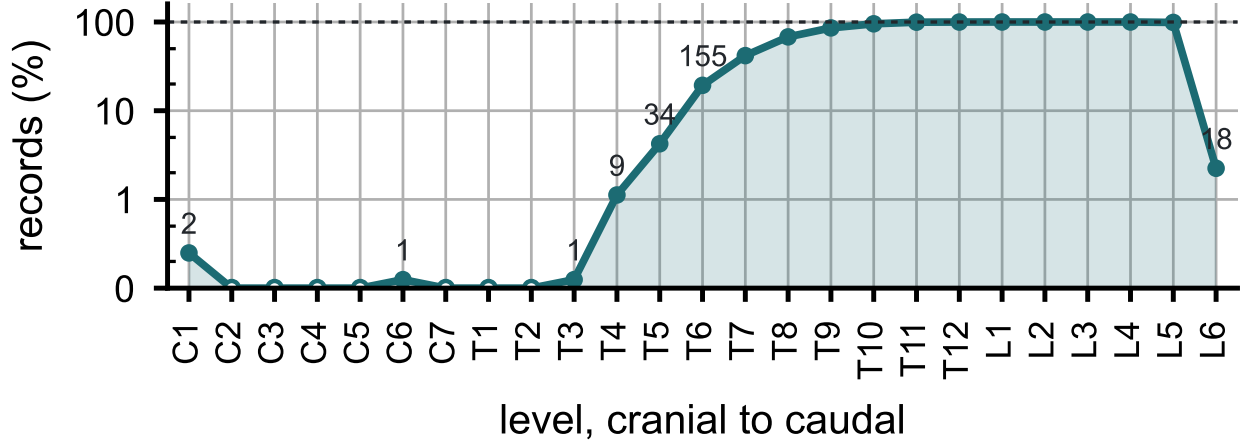


FIG. 7 Records carrying each vertebral level, cranial to caudal. Counts per identifier in Table S1.

380 T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat.
 381 The cohort is a colorectal screening population aged 50 and over, so its distributions should
 382 not be read as representative of a surgical one.

383 *Declared but empty, and confirmed so.* The partial-annotation sentinel (255) is declared
 384 in the scheme and occurs in no record, established by counting identifiers across all 802
 385 released volumes rather than asserted. It is retained because the protocol is part of the
 386 scheme and a later release may ship a partially traced record. The hardware identifiers were
 387 in this position until v6 and are no longer: 76–82 are populated in the eleven instrumented
 388 records described in Sec. II.I.

389 *Strength of the labels varies by structure, and the release does not average over that.*
 390 Vertebral labels derive from radiologist-supervised source annotations, corrected where those
 391 sources were wrong. Pelvic labels on records that lacked one are pseudolabeled. The rib
 392 layer is a pseudolabel whose human review was triaged by an automated rule rather than
 393 exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional
 394 labels derive from two independent sources and are not uniformly adjudicated, which is
 395 precisely why the measures reported here are count-free, and the Castellvi grades are a
 396 consensus of two radiology resident physicians. The sacral sub-division inherits its level
 397 from an automatic method, and in 100 records that level places S1 above half the sacrum
 398 or below fifteen percent of it, which no first sacral segment is; those records are flagged in
 399 the released quality-control table and should be filtered before any measurement depending
 400 on the sacral endplate.

401 *Structures are not universally present.* Sacrum, hips and femora are present in all 802
402 records; one record carries no S1, one lacks T12, and nine lack an L5 identifier, because
403 the structure lies outside the field of view or, for L5, because the lowest lumbar segment is
404 labeled L6 or incorporated into the sacrum. Users needing the caudal anchor should filter
405 on it explicitly.

406 VI. DISCUSSION

407 *Comparison with related material.* Against the collections in Table I, the contribution
408 is the crosswalk placing spine and pelvis on the same series, together with classes for the
409 anatomy an enumeration anomaly actually produces. VerSe^{21,22} comes closest in intent,
410 enriching for transitional anatomy and grading by Castellvi, but it does not segment the
411 sacrum, so the structure a grade describes has no mask; this release segments it, as L6 or as a
412 separately carved S1. What this release adds over TotalSegmentator²⁴ is not more anatomy
413 per scan but the classes an anomaly needs to be recorded accurately.

414 *Limitations.* The cohort comes from one acquisition protocol of a colorectal-screening
415 population aged 50 and over, so how well it generalizes to a pathologic population is untested.
416 The rib layer’s human review was triaged by an automated rule rather than performed on
417 every bone, so it guarantees only the absence of the failure modes that rule detects.

418 *Future directions.* The scheme has been extended in place more than once (femora and
419 S1, per-level ribs, lumbar ribs, hardware); imaging beyond this protocol, and the C2 anchor
420 it cannot supply, are the natural next additions.

421 VII. CONCLUSION

422 CTSpinoPelvic1K places spine, pelvis, per-level ribs and femora on one coordinate frame
423 in 802 records, and gives the anatomy that makes lumbar numbering ambiguous a class of
424 its own rather than recording it as something it is not. Because both counting anchors are
425 explicit, a level can be identified from a field of view that cannot support the conventional
426 count downward from C2, which is every record here.

427 DATA AVAILABILITY

428 The release described here is v8, permanently archived at <https://doi.org/10.5281/zenodo.22293878>; the concept DOI <https://doi.org/10.5281/zenodo.22139642> resolves
429 to the current version. The archive carries the loader, the label scheme, the quality-control
430 tables and the analysis code under an open-source licence. The repository URL identifies
431 the authors; it is on the title page and will be restored here for the camera-ready version.
432

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436 CONFLICT OF INTEREST

437 The authors have no relevant conflicts of interest to disclose.

438 ETHICS

439 This work uses publicly available, de-identified imaging and annotations derived from it.
440 The authors' institutional review board determined that it is not human participant research
441 under 45 CFR 46 and requires no oversight.

442 REFERENCES

- 443 ¹J. Lian, N. Levine, and W. Cho, A review of lumbosacral transitional vertebrae and
444 associated vertebral numeration, *Eur. Spine J.* **27**, 995–1004 (2018), doi:10.1007/s00586-
445 018-5554-8.
- 446 ²Expert Panel on Neurological Imaging, T. A. Hutchins, M. Peckham, et al., ACR Ap-
447 propriateness Criteria low back pain: 2021 update, *J. Am. Coll. Radiol.* **18**, S361–S379
448 (2021), doi:10.1016/j.jacr.2021.08.002.

- ³American College of Radiology, ACR–ASNR–SABI–SSR practice parameter for the performance of magnetic resonance imaging (MRI) of the adult spine, revised 2023 (Resolution 6), American College of Radiology, Reston, VA, 2023.
- ⁴G. P. Konin and D. M. Walz, Lumbosacral transitional vertebrae: classification, imaging findings, and clinical relevance, *AJNR Am. J. Neuroradiol.* **31**, 1778–1786 (2010), doi:10.3174/ajnr.A2036.
- ⁵Y. Deng et al., CTSpine1K: a large-scale dataset for spinal vertebrae segmentation in computed tomography, *Mach. Learn. Biomed. Imaging* **3**, 824–832 (2025), doi:10.59275/j.melba.2025-gf84.
- ⁶P. Liu et al., Deep learning to segment pelvic bones: large-scale CT datasets and baseline models, *Int. J. Comput. Assist. Radiol. Surg.* **16**, 749–756 (2021), doi:10.1007/s11548-021-02363-8.
- ⁷H. Möller, R. Graf, J. Schmitt, et al., SPINEPS: automatic whole spine segmentation of T2-weighted MR images using a two-phase approach to multi-class semantic and instance segmentation, *Eur. Radiol.* **35**, 1178–1189 (2025), doi:10.1007/s00330-024-11155-y.
- ⁸H. Möller, H. Schoen, R. Graf, et al., VERIDAH: solving enumeration anomaly aware vertebra labeling across imaging sequences, 2026, arXiv:2601.14066. doi:10.48550/arXiv.2601.14066.
- ⁹Y. Otake, S. Schafer, J. W. Stayman, et al., Automatic localization of vertebral levels in x-ray fluoroscopy using 3D-2D registration: a tool to reduce wrong-site surgery, *Phys. Med. Biol.* **57**, 5485–5508 (2012), doi:10.1088/0031-9155/57/17/5485.
- ¹⁰S.-F. L. Lo, Y. Otake, V. Puvanesarajah, et al., Automatic localization of target vertebrae in spine surgery: clinical evaluation of the LevelCheck registration algorithm, *Spine (Phila. Pa. 1976)* **40**, E476–E483 (2015), doi:10.1097/BRS.0000000000000814.
- ¹¹T. De Silva, S.-F. L. Lo, N. Aygun, et al., Utility of the LevelCheck algorithm for decision support in vertebral localization, *Spine (Phila. Pa. 1976)* **41**, E1249–E1256 (2016), doi:10.1097/BRS.0000000000001589.
- ¹²D. S. Kreiner, W. O. Shaffer, J. L. Baisden, et al., An evidence-based clinical guideline for the diagnosis and treatment of degenerative lumbar spinal stenosis (update), *Spine J.* **13**, 734–743 (2013), doi:10.1016/j.spinee.2012.11.059.
- ¹³American College of Surgeons Committee on Trauma, ACS TQIP best practices guidelines in imaging, American College of Surgeons, Chicago, 2018.

¹⁴K. Smith et al., Data from CT COLONOGRAPHY, The Cancer Imaging Archive, 2015,
doi:10.7937/K9/TCIA.2015.NWTESAY1.

¹⁵K. Clark et al., The Cancer Imaging Archive (TCIA): maintaining and operating a public
information repository, *J. Digit. Imaging* **26**, 1045–1057 (2013), doi:10.1007/s10278-013-
9622-7.

¹⁶M. R. Zindrick et al., Analysis of the morphometric characteristics of the thoracic and
lumbar pedicles, *Spine (Phila. Pa. 1976)* **12**, 160–166 (1987), doi:10.1097/00007632-
198703000-00012.

¹⁷M. M. Panjabi et al., Thoracic human vertebrae: quantitative three-dimensional anatomy,
Spine (Phila. Pa. 1976) **16**, 888–901 (1991), doi:10.1097/00007632-199108000-00006.

¹⁸M. M. Panjabi et al., Human lumbar vertebrae: quantitative three-dimensional anatomy,
Spine (Phila. Pa. 1976) **17**, 299–306 (1992), doi:10.1097/00007632-199203000-00010.

¹⁹E. C. Benzel, *Biomechanics of Spine Stabilization*, Thieme, New York, 3rd edition, 2015.

²⁰R. J. Hughes and A. Saifuddin, Numbering of lumbosacral transitional vertebrae on
MRI: role of the iliolumbar ligaments, *AJR Am. J. Roentgenol.* **187**, W59–W65 (2006),
doi:10.2214/AJR.05.0415.

²¹H. Liebl et al., A computed tomography vertebral segmentation dataset with anatomical
variations and multi-vendor scanner data, *Sci. Data* **8**, 284 (2021), doi:10.1038/s41597-
021-01060-0.

²²A. Sekuboyina et al., VerSe: a vertebrae labelling and segmentation bench-
mark for multi-detector CT images, *Med. Image Anal.* **73**, 102166 (2021),
doi:10.1016/j.media.2021.102166.

²³L. Jin et al., RibSeg v2: a large-scale benchmark for rib labeling and
anatomical centerline extraction, *IEEE Trans. Med. Imaging* **43**, 570–581 (2024),
doi:10.1109/TMI.2023.3313627.

²⁴J. Wasserthal et al., TotalSegmentator: robust segmentation of 104 anatomic structures
in CT images, *Radiol. Artif. Intell.* **5**, e230024 (2023), doi:10.1148/ryai.230024.

²⁵H. Möller et al., Automated thoracolumbar stump rib detection and analysis in a large
CT cohort, 2025, arXiv:2505.05004. doi:10.48550/arXiv.2505.05004.

²⁶[Reference to the authors’ own work, removed for double-anonymized review.]

²⁷F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, nnU-Net:
a self-configuring method for deep learning-based biomedical image segmentation, *Nat.*

513 Methods **18**, 203–211 (2021), doi:10.1038/s41592-020-01008-z.

514 ²⁸R. Vialle, N. Levassor, L. Rillardon, A. Tempier, W. Skalli, and P. Guigui, Radiographic
515 analysis of the sagittal alignment and balance of the spine in asymptomatic subjects, J.
516 Bone Joint Surg. Am. **87**, 260–267 (2005), doi:10.2106/JBJS.D.02043.

517 ²⁹A. E. Castellvi, L. A. Goldstein, and D. P. K. Chan, Lumbosacral transitional vertebrae
518 and their relationship with lumbar extradural defects, Spine (Phila. Pa. 1976) **9**, 493–495
519 (1984), doi:10.1097/00007632-198407000-00014.

520 ³⁰P. J. Pickhardt, B. D. Pooler, T. Lauder, A. M. del Rio, R. J. Bruce, and N. Binkley,
521 Opportunistic screening for osteoporosis using abdominal computed tomography scans
522 obtained for other indications, Ann. Intern. Med. **158**, 588–595 (2013), doi:10.7326/0003-
523 4819-158-8-201304160-00003.

524 ³¹N. E. Epstein, A perspective on wrong level, wrong side, and wrong site spine surgery,
525 Surg. Neurol. Int. **12**, 286 (2021), doi:10.25259/SNI.402_2021.

526 ³²E. Beaulieu et al., A review of finite element analysis in spine surgery decision-making, J.
527 Clin. Med. **15**, 2584 (2026), doi:10.3390/jcm15072584.

528 FIGURE CAPTIONS

529 **Figure 1.** How the dataset was built. n counts scans, or corrections, at each step.

530 **Figure 2.** The two anchors, from the released labels. Red, lowest rib-bearing vertebra and
531 its rib; blue, S1; yellow, the rib-free vertebrae between. (b) and (c) both count four.

532 **Figure 3.** Instrumentation in the release, one exemplar each, drawn through bone. (a)
533 Hip arthroplasty, 80, $n = 8$. (b) Femoral-neck screws, 82, $n = 1$. (c) Sacroiliac screws, 81,
534 $n = 1$. (d) Interbody cages, 77, $n = 1$. Bar, 5 cm.

535 **Figure 4.** Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b)
536 Lowest rib length over the rib above. (c) Lowest lumbar transverse span against its gap to
537 the ala; transitional labels marked.

538 **Figure 5.** Spinopelvic measures against standing reference bands (mean $\pm$ SD)²⁸.

539 **Figure 6.** Morphometry by level: median, interquartile range, 5th–95th percentile, n at
540 right. (a) Body height. (b) Endplate width. (c) Canal. (d) Pedicle width. (e) Disc height.
541 (f) Trabecular attenuation.

542 **Figure 7.** Records carrying each vertebral level, cranial to caudal. Counts per identifier in

⁵⁴³ Table S1.